Gaussian Process Field-Level Inference of the Galaxy Density from Point Distributions using Scalable Vecchia Methods with 2D Nearest-Neighbour Diagnostics

A Collaboration Proposal

Tom Abel

Kavli Institute for Particle Astrophysics and Cosmology (KIPAC)

Stanford University

with contributions from P. Frank, B. Dodge, S. Clark, R. Wechsler, and collaborators

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Abstract

We propose a two-stage framework for cosmological inference from galaxy surveys that cleanly separates measurement from theoretical interpretation. **Stage 1** reconstructs the galaxy density field $\delta_{\rm g}(\boldsymbol{x})$ *directly at galaxy positions* using a scalable Gaussian process (GP) prior with a Poisson likelihood, and learns the anisotropic galaxy power spectrum $P_{gg,\ell}(k)$ as kernel hyperparameters—delivering band-power estimates with internally consistent covariances from the Fisher matrix of the GP marginal likelihood, without reference to the matter field, bias, or cosmology. The framework extends naturally to **multi-output GPs** that simultaneously learn the auto- and cross-power spectra of galaxy subsamples split by stellar mass, star formation rate, colour, or target class, enabling empirical measurement of property-dependent clustering and assembly bias. Because derivatives of a GP are themselves GPs, Stage 1 also delivers the **tidal shear tensor** and cosmic web classification at every galaxy position with propagated uncertainty, enabling empirical tests of whether galaxy properties depend on local anisotropy beyond their dependence on density—a direct, model-independent probe of tidal bias. **Stage 2** takes these spectra and fits any theoretical model (perturbative bias, HOD, EFTofLSS, simulation emulators) without re-running the GP inference. The `GraphGP` package—developed at Stanford—implements the Vecchia approximation with $\mathcal{O}(Nk^3)$ scaling, enabling Stage 1 on catalogs of $\sim 10^6$ galaxies. We propose to use 2D data–data k-th nearest neighbour (DD-kNN) statistics—measured at the same galaxy positions—as a Gaussianity diagnostic within Stage 1 and as a source of non-Gaussian constraints for Stage 2. We outline the framework, describe a validation program using AbacusSummit catalogs, and discuss the path to DESI data.

1 Motivation: Why Gaussian Processes?

The standard approach to galaxy survey analysis proceeds in two stages: (1) estimate summary statistics—power spectrum, correlation function, bispectrum, kNN-CDFs, etc.—from the catalog, and (2) compare these summaries to theoretical predictions to constrain cosmological parameters. This two-step procedure is well understood but discards information, and the estimation of summary statistics is entangled with assumptions about window functions, random catalogs, and binning choices.

We propose to replace the first stage with something fundamentally more principled: a Gaussian process reconstruction of the galaxy density field $\delta_{\rm g}$, evaluated directly at galaxy positions, that simultaneously delivers both the field and the optimal estimate of its covariance structure—the galaxy power spectrum $P_{gg}(k)$—with a full posterior covariance. The key insight is that this is naturally a **Gaussian process regression problem with a Poisson likelihood**, and that the GP kernel *is* the galaxy power spectrum.

Crucially, we propose to **separate the problem completely** into two stages:

- **Stage 1 (Empirical):** Reconstruct $\delta_{\rm g}$ and learn $P_{gg,\ell}(k)$ from the galaxy point pattern alone—no reference to the matter field, bias, or cosmology.
- **Stage 2 (Theoretical):** Take the inferred $P_{gg,\ell}(k_i)$ band-powers *with their full covariance matrix* and fit any theoretical model—perturbative bias expansion, HOD, simulation-based emulators, EFTofLSS—without re-running the GP inference.

This is directly analogous to how CMB analysis works: Stage 1 extracts C_ℓ with their covariance from the map (optimal map-making), Stage 2 fits cosmological parameters to C_ℓ. The GP here plays the role of the optimal power spectrum estimator—but for an irregular Poisson point process, not a pixelized field on a sphere.

2 Stage 1: The Galaxy Field as a Gaussian Process

2.1 The Generative Model

We model the observed galaxy catalog as a log-Gaussian Cox process. The generative model has two layers—note the deliberate absence of any matter field or bias model.

GP prior on the galaxy overdensity. The galaxy overdensity $\delta_{\rm g}$ is drawn from a GP with zero mean and covariance specified by the galaxy power spectrum:

$$\delta_{\rm g} \sim \mathcal{GP}\big(0,\ C_{gg}(\boldsymbol{x},\boldsymbol{x}')\big), \qquad C_{gg}(\boldsymbol{x},\boldsymbol{x}') = \int \frac{d^3k}{(2\pi)^3}\, P_{gg}(k)\, e^{i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')}\,. \tag{1}$$

The kernel $C_{gg}(r)$ is stationary and decaying—precisely the class for which the Vecchia approximation achieves linear scaling.

Here $P_{gg}(k)$ is not a theoretical prediction: it is a **free function to be learned from the data**. We parameterize it nonparametrically as a set of band-powers $\theta = \{P_{gg}(k_i)\}$ at discrete wavenumbers k_i, or in multipole form $\theta = \{P_{gg,\ell}(k_i)\}$ to capture anisotropy (Section 6).

Poisson likelihood. Given the field, galaxy positions follow an inhomogeneous Poisson process. The log-likelihood is:

$$\ln \mathcal{L}\big(\{\boldsymbol{x}_j\} \,|\, \delta_{\rm g}\big) = \sum_{j=1}^{N} \ln\big[\bar{n}\,(1+\delta_{\rm g}(\boldsymbol{x}_j))\big] - \int \bar{n}\,(1+\delta_{\rm g}(\boldsymbol{x}))\, d^3x\,. \tag{2}$$

This is purely a statement about how galaxies sample the galaxy field. No matter density, no bias, no cosmology.

2.2 The Posterior Over the Field

The posterior is:

$$p\big(\delta_{\rm g},\, \theta \,|\, \{\boldsymbol{x}_j\}\big) \;\propto\; \underbrace{\mathcal{L}_{\rm Poisson}\big(\{\boldsymbol{x}_j\}|\delta_{\rm g}\big)}_{\text{how galaxies sample the field}} \;\times\; \underbrace{p_{\rm GP}(\delta_{\rm g}|\theta)}_{\text{GP prior with kernel } C_{gg,\theta}} \;\times\; \underbrace{p(\theta)}_{\text{prior on band-powers}}\,, \tag{3}$$

where $\theta = \{P_{gg,\ell}(k_i)\}$. The GP prior term requires the inverse and log-determinant of $C_{gg,\theta}$—exactly what `GraphGP` provides in $\mathcal{O}(Nk^3)$ time.

2.3 Learning the Galaxy Power Spectrum: The Marginal Likelihood

The central object of Stage 1 is the GP marginal likelihood, obtained by integrating out the field:

$$p\big(\{\boldsymbol{x}_j\} \,|\, \theta\big) = \int p\big(\{\boldsymbol{x}_j\} \,|\, \delta_{\rm g}\big)\, p(\delta_{\rm g} \,|\, \theta)\; \mathcal{D}\delta_{\rm g}\,. \tag{4}$$

This is intractable for the Poisson likelihood but can be approximated via the Laplace method (finding the MAP field and expanding around it) or variational inference (NIFTy's MGVI). In either case, `GraphGP`'s differentiable (JAX) implementation of L^{-1} and $\ln\det L$ enables gradient-based optimization or HMC sampling of θ.

The galaxy power spectrum $P_{gg,\ell}(k_i)$ is thus learned as a set of kernel hyperparameters, optimized to maximize the probability of the observed galaxy point pattern under the Poisson-GP model. No FKP weights, no Landy–Szalay random catalogs, no window function deconvolution.

2.4 The Key Deliverable: Band-Powers with Their Covariance

Stage 1 delivers to Stage 2 exactly what is needed for standard cosmological parameter inference: a set of band-power estimates $\hat{P}_{gg,\ell}(k_i)$ together with their covariance matrix.

Fisher matrix from the marginal likelihood. At the MAP of the kernel hyperparameters, the Hessian of the log-marginal likelihood gives the Fisher information matrix:

$$F_{ij} = -\frac{\partial^2 \ln p(\{\boldsymbol{x}_j\} \,|\, \theta)}{\partial\theta_i\, \partial\theta_j}\bigg|_{\theta=\hat{\theta}}\,, \tag{5}$$

which is computable via JAX autodiff through `GraphGP`. The inverse F^{-1} is the covariance matrix of the band-power estimates.

Full posterior from HMC samples. If Stage 1 uses HMC rather than MAP optimization, the posterior samples $\{\theta^{(s)}\}_{s=1}^{S}$ over the band-powers capture any non-Gaussianity in the band-power posterior. These samples can be passed directly to Stage 2 as an implicit likelihood, no Gaussian approximation needed. This is strictly more informative than the Fisher matrix.

Comparison to traditional estimators. The FKP estimator (Feldman, Kaiser & Peacock, 1994) and Landy–Szalay estimator each deliver $\hat{P}(k_i)$ or $\hat{\xi}(r_i)$ with a covariance estimated from mocks or analytic approximations. The GP marginal likelihood delivers the same objects—but (i) handles irregular survey geometry exactly through the Poisson likelihood, (ii) produces internally consistent covariances from the curvature of the likelihood rather than from external mocks, and (iii) jointly delivers the reconstructed field.

2.5 Why the Vecchia Approximation Is Physically Motivated

The Vecchia approximation conditions each point only on its k nearest previously generated neighbors, inducing a sparse precision matrix. For cosmological fields with decaying correlations, this reflects genuine physics: knowing the field at nearby points screens distant ones. Katzfuss & Guinness (2021) showed that for Matérn-class kernels, the approximation error vanishes as k grows polylogarithmically in N.

3 Stage 2: Cosmological Interpretation

Stage 2 takes the output of Stage 1—band-powers $\hat{P}_{gg,\ell}(k_i)$ and their covariance $\hat{C}_{ij}$—and performs standard cosmological parameter inference.

3.1 The Stage 2 Likelihood

Given a theoretical model that predicts $P^{\rm th}_{gg,\ell}(k_i \,|\, \Omega_m, \sigma_8, b_1, b_2, \ldots)$, the Stage 2 likelihood is:

$$\ln \mathcal{L}_{\rm Stage\,2} = -\frac{1}{2}\,\Delta P_i\; F_{ij}\; \Delta P_j\,, \qquad \Delta P_i = \hat{P}_{gg,\ell}(k_i) - P^{\rm th}_{gg,\ell}(k_i)\,, \tag{6}$$

where F_{ij} is the Fisher matrix from Eq. (5). If HMC samples are available, the likelihood can be evaluated without a Gaussian assumption by importance-weighting the posterior samples.

3.2 Complete Freedom in the Theoretical Model

The power of the two-stage separation is that **any** theoretical model for P_{gg} can be plugged into Stage 2 without re-running Stage 1:

- **Perturbative bias:** $P_{gg}(k) = b_1^2\, P_{mm}(k) + b_1 b_2\, P_{b_1 b_2}(k) + \ldots$ within the EFTofLSS or standard PT framework.
- **Simulation-based emulators:** Neural-net emulators trained on AbacusSummit or similar suites that predict P_{gg} as a function of cosmology and HOD parameters.

- **Halo model:** Decomposition into 1-halo and 2-halo terms with HOD-parameterized galaxy profiles.
- **Hybrid approaches:** HEFT-style models (Banerjee, Kokron & Abel, 2022) that use perturbation theory for the large-scale bias and simulations for the small-scale response.

Different groups can explore different theoretical models on the same Stage 1 output, enabling direct comparison of modelling approaches without any ambiguity about how the data were compressed.

3.3 Handling Degeneracies Between Bias and Cosmology

A well-known challenge in galaxy clustering analysis is the degeneracy between the amplitude of matter fluctuations (σ_8) and the linear bias (b_1): the galaxy power spectrum is sensitive to $b_1^2\,\sigma_8^2$ on large scales. In the two-stage framework, this degeneracy is handled entirely in Stage 2. Stage 1 measures P_{gg} agnostically; Stage 2 explores the (σ_8, b_1) degeneracy surface given whatever priors or external data are available (CMB, lensing, cluster counts). The anisotropic multipoles $P_{gg,2}(k)$ and $P_{gg,4}(k)$ partially break this degeneracy because the Kaiser factor $(b_1 + f\mu^2)^2$ makes the quadrupole-to-monopole ratio sensitive to f/b_1, which is independent of $\sigma_8\, b_1$.

3.4 The kNN-CDF as a Non-Gaussian Extension

The DD-kNN statistics (Section 8) provide information beyond P_{gg}—they are sensitive to all connected N-point functions. In the two-stage framework, DD-kNN residuals (observed minus GP-predicted) quantify the non-Gaussian information content of $\delta_{\rm g}$ that the Gaussian GP cannot capture. These residuals can be passed to Stage 2 as additional constraints, or used as a diagnostic to identify the scales below which the Gaussian approximation for $\delta_{\rm g}$ breaks down and higher-order statistics become essential.

4 Multi-Tracer Extension: Learning Galaxy Properties and Clustering Simultaneously

So far we have treated all galaxies as identical Poisson samples of a single scalar field $\delta_{\rm g}$. In reality, each galaxy carries a vector of measured properties—stellar mass M_*, star formation rate $\dot{M}_*$, metallicity Z, colour, morphology, spectral features—and these properties correlate with the galaxy's environment. The GP framework extends naturally to learn this relationship, staying entirely within Stage 1.

4.1 The Physical Picture

A massive, red, quiescent galaxy lives in a different environment than a low-mass, blue, star-forming galaxy. Their clustering amplitudes differ (different effective bias), and the degree to which their environments overlap carries physical information about the galaxy–halo connection. In conventional analyses, this is captured by splitting the sample into

subsets and measuring auto- and cross-correlations. The multi-output GP does this jointly and optimally.

4.2 Multi-Output GP: Matrix-Valued Kernels

Partition the galaxy catalog into T subsamples $A, B, C, \ldots$ by observable property (e.g., stellar mass bins, colour splits, SFR quartiles, or DESI target classes: LRG, ELG, QSO, BGS). Each subsample traces a different density field $\delta_g^A, \delta_g^B, \ldots$. The GP prior becomes matrix-valued:

$$\begin{pmatrix} \delta_g^A(\boldsymbol{x}) \\ \delta_g^B(\boldsymbol{x}) \\ \vdots \end{pmatrix} \sim \mathcal{GP}\left(\mathbf{0}, \begin{pmatrix} C^{AA}(\boldsymbol{x},\boldsymbol{x}') & C^{AB}(\boldsymbol{x},\boldsymbol{x}') & \cdots \\ C^{BA}(\boldsymbol{x},\boldsymbol{x}') & C^{BB}(\boldsymbol{x},\boldsymbol{x}') & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \right), \tag{7}$$

where each block has its own power spectrum:

$$C^{AB}(\boldsymbol{x},\boldsymbol{x}') = \int \frac{d^3k}{(2\pi)^3}\, P_{gg}^{AB}(k)\, e^{i\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{x}')}\,. \tag{8}$$

The auto-spectra $P_{gg}^{AA}(k)$, $P_{gg}^{BB}(k)$ describe how strongly each subsample clusters; the cross-spectra $P_{gg}^{AB}(k)$ describe how their environments overlap. For T subsamples, this is a $T \times T$ matrix of power spectra, all learned simultaneously as kernel hyperparameters.

Poisson likelihood per subsample. Each subsample has its own mean density $\bar{n}^A$ and selection function $w^A(\boldsymbol{x})$. The total log-likelihood factorizes:

$$\ln \mathcal{L} = \sum_A \left[\sum_{j\in A} \ln\left[\bar{n}^A\, w^A(\boldsymbol{x}_j)\,(1+\delta_g^A(\boldsymbol{x}_j))\right] - \int \bar{n}^A\, w^A(\boldsymbol{x})\,(1+\delta_g^A(\boldsymbol{x}))\, d^3x \right], \tag{9}$$

which is diagonal across subsamples given the fields—the coupling between subsamples enters entirely through the off-diagonal blocks of the GP prior.

Computational cost. If the T subsamples together contain N galaxies, the multi-output GP involves a $(T \times T)$ block kernel evaluated at N positions. The Vecchia approximation applies identically: each galaxy is conditioned on its k nearest neighbours (which may come from any subsample), giving $\mathcal{O}(N(Tk)^3)$ scaling in the worst case, or $\mathcal{O}(NT^3k^3)$ if the block structure is exploited. For $T \sim 3$–5 and $k \sim 10$, this remains tractable.

4.3 What the Multi-Tracer GP Delivers to Stage 2

Stage 1 now delivers a richer data product: the full matrix of auto- and cross-power spectra $\hat{P}_{gg,\ell}^{AB}(k_i)$ together with their joint covariance. Stage 2 can exploit this in several ways.

Breaking the σ_8–b_1 degeneracy. In a perturbative bias framework, the large-scale auto-spectra go as $(b_1^A)^2\, P_{mm}(k)$ and $(b_1^B)^2\, P_{mm}(k)$, while the cross-spectrum goes as $b_1^A\, b_1^B\, P_{mm}(k)$. From three measurements one obtains two bias values and $P_{mm}(k)$ separately—the σ_8–b_1 degeneracy is broken without any external data (lensing, CMB). This is the multi-tracer technique of Seljak (2009), but now the auto- and cross-spectra are measured jointly and optimally by the GP, with internally consistent covariances.

Stochasticity and decorrelation. The cross-correlation coefficient

$$r^{AB}(k) = \frac{P_{gg}^{AB}(k)}{\sqrt{P_{gg}^{AA}(k)\, P_{gg}^{BB}(k)}} \tag{10}$$

measures the degree to which subsamples A and B trace the same underlying field. Deviations from $r^{AB} = 1$ indicate stochastic bias: the galaxy–halo connection has scatter that differs between subsamples. This is a direct, empirical measurement of a key ingredient in bias models, delivered by Stage 1 without assuming any theoretical form.

Assembly bias as a measurable kernel feature. If the clustering of galaxies at fixed number density depends on a secondary property (stellar mass, colour, SFR) beyond what halo mass alone predicts, this is galaxy assembly bias. In the multi-output GP, it manifests as a property-dependence of $P_{gg}^{AA}(k)$ that cannot be explained by a single universal kernel scaled by a property-dependent amplitude. Stage 1 measures this dependence; Stage 2 tests whether it is consistent with predictions from abundance matching, hydrodynamical simulations, or specific HOD extensions. The GP framework turns assembly bias from a modelling assumption into an observable.

4.4 Continuous Property Labels

For large spectroscopic surveys where M_*, SFR, and metallicity are available for each galaxy, partitioning into discrete subsamples is somewhat arbitrary. A more elegant extension is to let the kernel depend continuously on the property labels:

$$C_{gg}(\boldsymbol{x}_i, \boldsymbol{x}_j \,|\, \boldsymbol{p}_i, \boldsymbol{p}_j) = C_{\rm spatial}(\boldsymbol{x}_i - \boldsymbol{x}_j) \;\times\; C_{\rm prop}(\boldsymbol{p}_i, \boldsymbol{p}_j)\,, \tag{11}$$

where $C_{\rm spatial}$ encodes the spatial clustering and $C_{\rm prop}$ is a kernel on property space that learns how galaxies with similar (or different) properties co-cluster. This separable (product) form is the simplest ansatz; more general forms can be explored. The key point is that the GP learns $C_{\rm prop}$ from the data, revealing empirically which galaxy properties most strongly predict clustering environment—a question at the heart of galaxy formation physics.

4.5 Cross-kNN Statistics for Multi-Tracer Diagnostics

The DD-kNN extends naturally to the multi-tracer setting. For each galaxy in subsample A, find its k-th nearest neighbour in subsample B and record the distance decomposed into $(r_{\rm p}, r_\pi)$. The resulting 2D cross-kNN-CDF probes how different galaxy populations cluster

relative to each other, sensitive to all orders of cross-correlations (Banerjee & Abel, 2021b). In the GP framework, the predicted cross-kNN follows from the off-diagonal kernel block C^{AB}, and residuals diagnose whether the Gaussian approximation captures the cross-clustering or whether higher-order cross-correlations are present.

4.6 Natural Accommodation of DESI Target Classes

DESI observes four main galaxy target classes—LRGs, ELGs, QSOs, and BGS—that span very different environments, redshift ranges, and selection functions. Rather than analyzing each class separately and then attempting an *a posteriori* combination, the multi-output GP analyzes them jointly from the start. Each class has its own selection function $w^A(\boldsymbol{x})$ in the Poisson likelihood, naturally handling the different completeness. The cross-spectra between classes encode how the different populations overlap in the cosmic web, providing constraints on relative bias that are inaccessible to single-tracer analyses.

5 Local Environment from GP Derivatives: Tidal Tensor and Cosmic Web Classification

The multi-output GP of the previous section characterizes how galaxy properties relate to the *amplitude* of the local density field. But galaxies live in a cosmic web whose local geometry—node, filament, sheet, or void—is not captured by $\delta_{\rm g}$ alone. A galaxy at $\delta_{\rm g} = 0.5$ in a filament may have very different properties from one at $\delta_{\rm g} = 0.5$ in a collapsing node. To test whether galaxy properties depend on more than local density, we need local measures of *anisotropy*.

The GP framework provides these for free, with full uncertainty propagation, through a fundamental property of Gaussian processes: **derivatives of a GP are themselves GPs**.

5.1 Derivatives of Gaussian Processes

If $\delta_{\rm g} \sim \mathcal{GP}(0, C_{gg})$, then the field, its gradient, and its Hessian at any point are jointly Gaussian:

$$\begin{pmatrix} \delta_{\rm g}(\boldsymbol{x}) \\ \partial_i\, \delta_{\rm g}(\boldsymbol{x}) \\ \partial_i\partial_j\, \delta_{\rm g}(\boldsymbol{x}) \end{pmatrix} \sim \mathcal{GP}\left(\mathbf{0},\, \begin{pmatrix} C & \partial_{i'}C & \partial_{i'}\partial_{j'}C \\ \partial_i C & \partial_i\partial_{i'}C & \cdots \\ \partial_i\partial_j C & \cdots & \partial_i\partial_j\partial_{i'}\partial_{j'}C \end{pmatrix}\right), \tag{12}$$

where all entries are derivatives of the original kernel C_{gg} with respect to its arguments, and primed indices refer to $\boldsymbol{x}'$. No additional model or computation is needed: the derivative kernels follow analytically from C_{gg}.

The GP posterior therefore delivers, *at each galaxy position*, not just $\delta_{\rm g}(\boldsymbol{x}_i)$ with uncertainty, but also the gradient $\nabla\delta_{\rm g}$ and the full symmetric Hessian $\mathcal{H}_{ij} = \partial_i\partial_j\delta_{\rm g}$—a vector of $1 + 3 + 6 = 10$ quantities per galaxy, all with joint posterior uncertainty propagated from the data. `GraphGP` explicitly supports derivatives through custom JAX primitives, so this computation is integrated into the existing infrastructure.

5.2 The Tidal Tensor and Cosmic Web Classification

The Hessian of the density field decomposes into a trace (the Laplacian $\nabla^2\delta_{\rm g}$, related to curvature and convergence of flows) and a traceless symmetric part, the *tidal shear tensor*:

$$s_{ij} = \left(\frac{\partial_i\partial_j}{\nabla^2} - \frac{\delta^K_{ij}}{3}\right)\delta_{\rm g}\,, \tag{13}$$

where δ^K_{ij} is the Kronecker delta. The eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$ of the Hessian (or equivalently of s_{ij} plus the trace part) classify the local cosmic web environment:

Environment	λ_1	λ_2	λ_3
Node (cluster)	$+$	$+$	$+$
Filament	$+$	$+$	$-$
Sheet (wall)	$+$	$-$	$-$
Void	$-$	$-$	$-$

This classification comes directly from the GP posterior at every galaxy position—with no separate classification algorithm, no arbitrary smoothing scale, and with fully propagated uncertainty on the eigenvalues.

5.3 Why Not Bond-Orientational Order Parameters?

An alternative approach to local anisotropy is the Steinhardt–Nelson–Ronchetti bond-orientational order parameters q_l, defined as angular averages of spherical harmonics over nearest-neighbour bonds:

$$q_{lm}(\boldsymbol{x}_i) = \frac{1}{N_b}\sum_{j=1}^{N_b} Y_{lm}(\hat{r}_{ij})\,. \tag{14}$$

These are well-established in condensed matter physics for characterizing local crystalline order. However, they are fundamentally statistics of the *discrete point process*, not of the continuous field. They mix together the anisotropy of the underlying field with Poisson shot noise and 1-halo effects (the angular distribution of satellite galaxies within a halo). They have no natural relationship to the GP, and their uncertainty must be estimated separately. They remain useful as a complementary diagnostic on the discrete galaxy positions—analogous to the DD-kNN but sensitive to angular rather than radial structure—but they are not the primary tool for characterizing field-level anisotropy.

The GP tidal tensor, by contrast, is a property of the continuous field $\delta_{\rm g}$, derived within the same probabilistic model that reconstructs the field, with jointly estimated uncertainty. It connects directly to the theoretical bias expansion (see below) and cleanly separates field-level anisotropy from shot-noise effects.

5.4 Connection to the Bias Expansion

The tidal tensor is not merely a diagnostic—it is the central object in the Lagrangian bias expansion used in Stage 2. The galaxy overdensity is expanded in terms of invariants of the

local gravitational environment:

$$\delta_{\rm g} = b_1\,\delta + \frac{b_2}{2}\,\delta^2 + b_{s^2}\,s^2 + b_{\nabla^2\delta}\,\nabla^2\delta + \dots\,, \tag{15}$$

where $s^2 = s_{ij}\,s_{ij}$ is the tidal shear scalar and the b coefficients are bias parameters. In our framework:

- Stage 1 delivers the GP posterior over $\delta_{\rm g}$, $\nabla^2\delta_{\rm g}$, and $s_{ij}^{(g)}$ at every galaxy position.
- These are properties of the *galaxy* field. At leading order, $s_{ij}^{(g)} \approx b_1\,s_{ij}^{(m)}$, so the cosmic web classification based on $\delta_{\rm g}$ is the same as that based on $\delta_{\rm m}$, up to a linear rescaling.
- Stage 2 then tests whether the observed correlations between galaxy properties and $(s^2, \nabla^2\delta)$ are consistent with the bias expansion, mapping from galaxy-frame to matter-frame quantities via the bias parameters.

The key point is that **Stage 1 measures the dependence of galaxy properties on local tidal environment empirically, without assuming any bias model**. Stage 2 interprets the measurements.

5.5 Enriching the Multi-Output GP with Tidal Labels

The tidal eigenvalues $(\lambda_1, \lambda_2, \lambda_3)$ and invariants $(s^2, \nabla^2\delta_{\rm g})$ become galaxy-level labels, on the same footing as stellar mass, SFR, or metallicity. They can be incorporated into the multi-output GP framework (Section 4) in two ways.

Conditional splitting. Partition galaxies not only by observable properties but also by tidal environment (e.g., filament vs. void, or quartiles of s^2). The multi-output GP then learns $P_{gg}^{AB}(k)$ where A and B are joint bins of (stellar mass $\times$ tidal environment). This directly answers: does the clustering of massive galaxies differ between filaments and voids, beyond what $\delta_{\rm g}$ alone predicts?

Continuous kernel conditioning. Extend the property-dependent kernel (Section 4) to include tidal labels:

$$C_{gg}(\boldsymbol{x}_i, \boldsymbol{x}_j \,|\, \boldsymbol{p}_i, \boldsymbol{p}_j,\, \boldsymbol{\lambda}_i, \boldsymbol{\lambda}_j) = C_{\rm spatial}(\boldsymbol{x}_i - \boldsymbol{x}_j)\;\times\; C_{\rm prop}(\boldsymbol{p}_i, \boldsymbol{p}_j)\;\times\; C_{\rm tidal}(\boldsymbol{\lambda}_i, \boldsymbol{\lambda}_j)\,, \tag{16}$$

where $C_{\rm tidal}$ learns how galaxies in similar (or different) tidal environments co-cluster. If $C_{\rm tidal}$ is non-trivial (i.e., not proportional to the identity), tidal bias is present.

5.6 Testing the Completeness of the Bias Expansion

The GP derivative framework enables a systematic, empirical test of whether the bias expansion of Eq. (15) is complete on a given set of scales. The logic is as follows:

1. If galaxy properties correlate with $\delta_{\rm g}$ alone, only b_1 (and possibly b_2) are needed.
2. If, at fixed $\delta_{\rm g}$, galaxy properties additionally correlate with s^2, then tidal bias b_{s^2} is present.
3. If, at fixed $(\delta_{\rm g}, s^2)$, they additionally correlate with $\nabla^2\delta_{\rm g}$, higher-derivative bias $b_{\nabla^2\delta}$ is required.

4. If residual correlations persist with the *eigenvalue ratios* (e.g., prolateness λ_1/λ_3) at fixed invariants, the bias expansion truncated at second order is insufficient.

Each step is a conditional correlation test on GP posterior samples, with uncertainty propagated from the data. This turns the question of "how many bias parameters do we need?" from a theoretical assumption into an empirical measurement—at each scale, for each galaxy population, in Stage 1.

5.7 Relation to Void and Filament Statistics

The GP-derived tidal classification subsumes several statistics that are currently computed independently: void catalogs (regions where all $\lambda_i < 0$), filament catalogs (from the eigenvector field), and the density-split statistics that condition clustering measurements on the local density. In the GP framework, these are all projections of the same posterior over $(\delta_{\rm g}, \nabla\delta_{\rm g}, \mathcal{H}_{ij})$, computed once and used many times. The posterior uncertainty on the classification propagates naturally: a galaxy at the filament/sheet boundary has large uncertainty in its classification, and this uncertainty is accounted for in any downstream analysis.

6 Anisotropic Kernels and Redshift-Space Distortions

Nothing about the Vecchia approximation requires an isotropic kernel. In redshift space, peculiar velocities distort the galaxy clustering along the line of sight. The redshift-space galaxy power spectrum depends on the angle $\mu = \hat{k} \cdot \hat{z}$:

$$P^s_{gg}(k,\mu) = \left(b_1 + f\mu^2\right)^2 P_m(k) \times D_{\rm FoG}(k\mu\sigma_v)\,. \tag{17}$$

In Stage 1, we do not decompose P^s_{gg} into bias and matter components. Instead, we parameterize it directly via its Legendre multipoles $P_{gg,\ell}(k)$, which encode the full anisotropic structure as free functions.

The corresponding real-space kernel becomes anisotropic:

$$C_{gg}(r_{\rm p}, r_\parallel) = \sum_{\ell=0,2,4} \int \frac{k^2\,dk}{2\pi^2}\; P_{gg,\ell}(k)\; j_\ell(kr)\; \mathcal{L}_\ell\!\left(\frac{r_\parallel}{r}\right), \tag{18}$$

where j_ℓ are spherical Bessel functions and $\mathcal{L}_\ell$ are Legendre polynomials. This kernel is still stationary and decaying—exactly the class `GraphGP` targets. The only modification is a two-dimensional kernel lookup table $C_{gg}(r_{\rm p}, r_\parallel)$ replacing the one-dimensional $C_{gg}(r)$.

The multipole moments of P_{gg} are literally the angular moments of the GP kernel, and **learning the kernel is equivalent to measuring the multipoles**—but at the field level. In Stage 1, these are free functions. In Stage 2, the decomposition into f, b_1, Alcock-Paczyński distortions, and cosmological parameters is performed on the inferred $P_{gg,\ell}(k_i)$ with their covariance.

7 A Pedagogical Introduction to kNN Statistics

The k-th nearest neighbour cumulative distribution function (kNN-CDF), introduced by Banerjee & Abel (2021a), is a set of summary statistics for discrete point distributions. We

describe them here from first principles, as they are central to this proposal.

7.1 Basic Construction

Consider N galaxies ("data points") distributed in a volume V. The kNN-CDF answers the following question: *if I pick a random point in space, what is the probability that its k-th nearest galaxy lies within distance r?*

Operationally, one places a dense set of $N_R \gg N$ uniformly distributed "query" points throughout the volume. For each query point q_j, one finds the distance $r_j^{(k)}$ to its k-th nearest galaxy. The empirical CDF of the set $\{r_j^{(k)}\}_{j=1}^{N_R}$ is the kNN-CDF for order k. The ensemble of these CDFs for $k = 1, 2, 3, \ldots$ forms the full kNN summary.

7.2 Why kNN-CDFs Capture All N-Point Functions

The mathematical power of kNN statistics lies in their connection to the generating function formalism. Banerjee & Abel (2021a) showed that the kNN-CDF at order k is related to volume integrals of all connected N-point correlation functions ξ_N of the underlying density field, up to and including order $N = k$. Specifically, the probability that a sphere of radius r centered on a random point contains *fewer than* k galaxies depends on ξ_1 (the mean density), ξ_2 (the two-point function), ξ_3 (the three-point function), and so on. Since a single measurement sweep through $k = 1, 2, \ldots, K$ captures information from all N-point functions up to order K, the kNN-CDFs are a highly efficient compression of the full clustering information.

7.3 Two Flavours: RD-kNN and DD-kNN

Yuan, Zamora & Abel (2023) introduced a key distinction between two flavours:

RD-kNN (Random–Data). The query points are volume-filling randoms. This probes the *density field*: how galaxies fill space around randomly chosen locations. It is sensitive to counts-in-cells and the void probability function.

DD-kNN (Data–Data). The query points are the galaxies themselves: for each galaxy, find its k-th nearest *other* galaxy. This probes the *clustering*: the local environment around galaxies, the group and cluster structure, and the phase-space topology. The DD-kNN is a strict generalization of the two-point correlation function: the 2PCF can be derived from the DD-kNN by marginalization, but the DD-kNN retains additional information about the ordering of neighbours at each distance (Yuan, Zamora & Abel, 2023).

7.4 Generalization to 2D: Capturing Redshift-Space Distortions

Yuan, Zamora & Abel (2023) further generalized the kNN to two dimensions by decomposing the distance to the k-th nearest neighbor into components transverse (r_p) and along (r_π) the line of sight:

$$r \longrightarrow (r_\mathrm{p}, r_\pi)\,. \tag{19}$$

The 2D kNN-CDF is then a function of both r_p and r_π: $F_k(r_\mathrm{p}, r_\pi)$ = fraction of query points whose k-th nearest data point lies within (r_p, r_π). The 2D CDF is computed by accumulating counts from all bins with smaller r_p *and* smaller r_π.

The 2D structure encodes redshift-space distortions directly: the Kaiser squashing at large separations and the Fingers-of-God elongation at small separations appear as distortions of the CDF contours away from the circular contours expected for an isotropic field (see Figures 2 and 3 of Yuan, Abel & Wechsler 2024).

8 DD-kNN as the Natural Observable for GP Inference

The 2D DD-kNN is the ideal observable for the GP framework proposed here, for several reinforcing reasons.

8.1 Shared Support: Both Live at Galaxy Positions

The DD-kNN is evaluated at the galaxy positions themselves—the same points where the GP field $\delta_\mathrm{g}(\boldsymbol{x}_i)$ is defined. Observable and model share the same support. No random catalog, no grid, no interpolation.

8.2 The Anisotropic Kernel Predicts the 2D DD-kNN Contours

The distortions of the 2D DD-kNN contours in the (r_p, r_π) plane are direct imprints of the anisotropic kernel $C(r_\mathrm{p}, r_\parallel)$. The Kaiser squashing and Fingers-of-God elongation visible in the CDF contours are precisely the angular structure of the kernel multipoles $C_\ell(r)$. Learning the kernel is equivalent to fitting the 2D kNN-CDF.

8.3 Sensitivity Beyond Two Points: A Built-In Model Adequacy Check

The GP kernel is a two-point object: it specifies the covariance of the field. If the GP model is correct (Gaussian field with the right bias model), then the DD-kNN predicted from the GP posterior should match the observed DD-kNN *at all k orders*. If it does not, the residuals diagnose what is wrong:

- $k = 1$–2: sensitive to the smallest-scale clustering and highest-density peaks (1-halo regime);
- $k = 3$–5: probes the transition between 1-halo and 2-halo regimes;
- $k > 5$: probes progressively larger scales and lower-density environments.

Non-Gaussianity of the field, or bias model inadequacy, will manifest as k-dependent residuals—a diagnostic stratified by scale and density environment that no two-point statistic can provide.

8.4 Incorporating DD-kNN into the Inference

The DD-kNN can enter Stage 1 in two complementary ways. First, as a *posterior predictive check*: given the GP posterior over $\delta_{\rm g}$, draw field samples, generate Cox process realizations, compute the predicted 2D DD-kNN CDF, and compare to the observed CDF. Agreement validates the Gaussian assumption for $\delta_{\rm g}$ on the scales probed; disagreement quantifies the non-Gaussian information content that the GP kernel misses. Second, more ambitiously, as an *additional likelihood term* within Stage 1:

$$\ln p(\theta|\text{data}) = \underbrace{\ln \mathcal{L}_{\text{Poisson}}}_{\text{field-level}} + \underbrace{\ln \mathcal{L}_{\text{DD-kNN}}}_{\text{summary-level}} + \ln p_{\text{GP}}(\delta_{\rm g}|\theta) + \ln p(\theta)\,. \tag{20}$$

The Poisson term constrains the field; the DD-kNN term constrains the kernel shape and tests for non-Gaussianity. Following Yuan, Abel & Wechsler (2024), the DD-kNN likelihood can be taken as Gaussian with a covariance estimated from jackknife resampling:

$$\ln \mathcal{L}_{\text{DD-kNN}} = -\frac{1}{2}(\boldsymbol{x}_{\text{pred}} - \boldsymbol{x}_{\text{obs}})^\top C_{\text{kNN}}^{-1}(\boldsymbol{x}_{\text{pred}} - \boldsymbol{x}_{\text{obs}})\,, \tag{21}$$

where $\boldsymbol{x}$ is the flattened DD-kNN data vector over k, $r_{\rm p}$, and r_π bins.

8.5 Cross-Validation: RD-kNN Predicted for Free

The cross-validation strategy of Yuan, Abel & Wechsler (2024) translates directly into the GP framework. The GP posterior mean field is a density reconstruction, and the RD-kNN (a density statistic) is predicted from it automatically. If the model is fit to the DD-kNN (a clustering statistic) and the predicted RD-kNN matches the observation, the model passes a stringent self-consistency test probing disjoint aspects of the galaxy field. This cross-validation comes for free in the GP framework.

9 Connection to the Wiener Filter

In the Gaussian-likelihood limit, the GP posterior mean reduces to the Wiener filter:

$$\hat{\delta}_{\text{WF}} = C_{gg}\,(C_{gg} + \sigma_n^2 I)^{-1}\,\boldsymbol{d}\,, \tag{22}$$

which is the workhorse of BORG-like codes at initialization. Our Stage 1 generalizes it in three directions: (i) Poisson replaces Gaussian noise, (ii) the field lives at irregular galaxy positions rather than on a grid, and (iii) the kernel C_{gg} is learned rather than fixed. The separation from Stage 2 means the Wiener filter need not assume any relationship between $\delta_{\rm g}$ and $\delta_{\rm m}$.

10 Existing Work and the Gap

Grid-based field-level inference. The BORG algorithm (Jasche & Wandelt, 2013) samples initial conditions on a regular grid and evolves them with 2LPT or N-body, comparing to galaxy counts via a Poisson likelihood. Seljak et al. (2017) showed that MAP estimation with a Gaussian likelihood approximation is much cheaper. All these approaches work on grids, creating difficulties with survey masks and irregular boundaries.

Scalable GPs on irregular points. Edenhofer et al. (2022) introduced Iterative Charted Refinement (ICR), an $\mathcal{O}(N)$ GP method applied spectacularly to 3D Galactic dust mapping with 661 million DOF (Edenhofer et al., 2024). `GraphGP` (Dodge, Frank & Clark, 2024) evolves ICR with a Vecchia approximation using k-nearest-neighbor graphs, adding exact inverse/determinant and CUDA extensions.

kNN-CDF statistics. Banerjee & Abel (2021a) introduced kNN-CDFs; Banerjee & Abel (2021b) extended them to cross-correlations; Banerjee, Kokron & Abel (2022) modelled them with HEFT; Yuan, Zamora & Abel (2023) generalized to 2D; and Yuan, Abel & Wechsler (2024) demonstrated robust cosmological inference from non-linear scales in the Beyond-2p blind challenge.

The gap. No existing work combines: (i) a scalable GP at galaxy positions with (ii) a Poisson likelihood and (iii) nonparametric learning of the anisotropic galaxy power spectrum $P_{gg,\ell}(k)$ as kernel hyperparameters, delivering band-powers with internally consistent covariances to (iv) a cleanly separated cosmological interpretation stage, using (v) kNN-CDFs as both observable and Gaussianity diagnostic.

11 Proposed Work

11.1 Phase 1: Stage 1 Proof of Concept on Simulation-Based Catalogs

We will validate Stage 1 using mock catalogs from the AbacusSummit simulation suite (Maksimova et al., 2021), which provides 85 cosmologies in $(2\,h^{-1}\,\mathrm{Gpc})^3$ volumes with high mass resolution.

1. **Mock catalog generation.** Populate AbacusSummit halos with galaxies using the AbacusHOD code (Yuan et al., 2021), applying redshift-space distortions along a chosen line of sight. The true $P_{gg,\ell}(k)$ is computed directly from the mock.
2. **GP prior construction.** Compute the anisotropic real-space kernel $C_{gg}(r_\mathrm{p}, r_\parallel)$ from an initial guess for $P_{gg,\ell}(k)$. Feed to `GraphGP` with the neighbor graph built using an anisotropic distance metric.
3. **Stage 1 inference.** Maximize the approximate marginal likelihood over band-power hyperparameters $\{P_{gg,\ell}(k_i)\}$. Compute the Fisher matrix (Eq. 5) to obtain the covariance of the band-power estimates. Compare inferred $P_{gg,\ell}(k_i)$ to the known truth from the mock.
4. **DD-kNN validation.** From the GP posterior field samples, generate Cox process realizations and compute predicted 2D DD-kNN CDFs. Compare to observed DD-kNN on the mock. Quantify the scales at which the Gaussian GP ceases to reproduce the observed DD-kNN, identifying where non-Gaussian information becomes important.
5. **Stage 2 cosmology recovery.** Pass the inferred $\hat{P}_{gg,\ell}(k_i)$ and F^{-1} to a standard Stage 2 likelihood. Fit cosmological and bias parameters using both a perturbative model and a simulation-based emulator. Verify unbiased recovery and compare constraining power to the direct emulator approach of Yuan, Abel & Wechsler (2024).

11.2 Phase 2: Stress-Testing the Two-Stage Separation

1. **Simulation code systematics.** The Beyond-2p blind challenge showed that simulation code, halo finder, and HOD implementation introduce features detectable in DD-kNN below $r_{\rm p} \sim 5\,h^{-1}\,{\rm Mpc}$ (Yuan, Abel & Wechsler, 2024). In the two-stage framework, these enter Stage 1 as features of P_{gg} that the GP kernel must absorb. We will test whether the nonparametric kernel learning can capture these features without introducing bias in the Stage 2 cosmological parameters.
2. **Information content of the Fisher matrix.** A central question is whether the GP-derived Fisher matrix F_{ij} for the band-powers is competitive with mock-based covariance estimates. We will compare the Stage 1 Fisher matrix to the sample covariance from 1000+ AbacusSummit realizations at fixed cosmology.
3. **Stage 2 model comparison.** Using the same Stage 1 output, fit three different Stage 2 models (perturbative bias, HOD emulator, HEFT) and verify that cosmological parameters are consistent. This tests whether the two-stage separation is clean enough that different theoretical models can be compared on equal footing.
4. **Non-Gaussianity frontier.** Use the DD-kNN residuals to map the scale at which the Gaussian assumption for $\delta_{\rm g}$ breaks down as a function of galaxy sample density and tracer type. This identifies the regime where Stage 1 can be trusted and where extensions (e.g., log-normal GP, higher-order corrections) become necessary.

11.3 Phase 3: Application to DESI Data

Applying this framework to actual survey data introduces several challenges that must be addressed:

Survey geometry and completeness. DESI's footprint, fiber assignment incompleteness, and bright-time/dark-time target selection create a spatially varying selection function $w(\boldsymbol{x})$. In the GP framework, this enters the Poisson rate as $\lambda_i = \bar{n}\, w(\boldsymbol{x}_i)\,(1+\delta_{\rm g}(\boldsymbol{x}_i))$ without any window function deconvolution—a significant advantage. However, the selection function must be accurately characterized, particularly at the boundary of tiles and in regions of fiber collision.

Fiber collisions. DESI's robotic fiber positioners cannot place two fibers closer than $\sim 60''$, creating missing pairs at small angular separations. For DD-kNN at low k orders and small $r_{\rm p}$, this is a direct source of systematic bias. Yuan, Hadzhiyska & Abel (2023) developed probabilistic redshift recovery techniques for correcting fiber collision effects on novel summary statistics. These corrections can be incorporated into the GP framework by modifying the Poisson likelihood to account for the probability that a true galaxy was missed.

Redshift evolution on the lightcone. DESI LRGs span $0.4 < z < 1.1$, over which the power spectrum, bias, and growth rate evolve. On a grid, this is handled by evolving the forward model through redshift slices. In the GP framework, a non-stationary kernel that

varies along the radial (redshift) direction is the natural description. The Vecchia approximation handles non-stationarity natively—it only requires that conditional correlations are local, which they remain on the lightcone.

Covariance estimation. The DD-kNN likelihood requires a covariance matrix. For the mock analyses, this is estimated from jackknife resampling (Yuan, Abel & Wechsler, 2024). For DESI, the large survey volume makes jackknife estimation feasible, but the covariance must account for the survey geometry. An alternative is to use the GP posterior samples themselves to estimate the covariance of predicted DD-kNN CDFs, providing an internally consistent error model.

Computational scaling. DESI will ultimately contain $\sim 10^7$ galaxy redshifts. While `GraphGP` scales to billions of points for generation, the iterative inference (HMC) requires many forward passes. For 10^6 galaxies this is tractable on a single A100; for 10^7 we may need to either subsample, work in redshift slices, or exploit the natural parallelism of the Vecchia factorization across GPUs.

12 Advantages Over Existing Approaches

Feature	**BORG**	**ICR/Dust**	**FKP/LS**	**This work**
At galaxy positions	no	yes†	no	yes
Poisson likelihood	yes	yes	no	yes
Learns P_{gg} from data	partial	no	yes‡	yes
Full posterior over field	yes	yes	no	yes
Anisotropic kernel (RSD)	yes	no	yes‡	yes
No window deconvolution	no	yes	no	yes
Internal covariances	no	partial	no	yes
Measurement/theory separated	no	n/a	yes	yes
Multi-tracer joint inference	no	no	separate	built-in
Tidal tensor / cosmic web	no	no	no	built-in
kNN diagnostic	no	no	separate	built-in
Scalable to 10^6+	yes	yes	yes	yes
Differentiable (JAX)	partial	yes	no	yes

†Applied to stars/dust, not galaxies. ‡Estimates $P(k)$ from pair counts, not field-level.

13 Team, Resources, and Invitation to Collaborate

This project sits at the intersection of several active research programs at Stanford and draws on a convergence of expertise that is, to our knowledge, unique. The people listed below have expressed interest in contributing, but this document is explicitly intended as an *open invitation*: anyone whose expertise touches on the methods or science described here is warmly encouraged to join.

Core expertise at Stanford. The key methodological ingredients—scalable Gaussian processes on irregular point distributions (`GraphGP`, ICR, NIFTy), cosmological simulations and large-scale structure theory, kNN-CDF statistics, and simulation-based galaxy–halo connection modelling—are all actively developed by researchers at KIPAC, the ISM group in Physics, and SLAC. Among those who have contributed to shaping this proposal:

- **Tom Abel** (KIPAC) — large-scale structure, kNN-CDF statistics, cosmological simulations.
- **Philipp Frank** (KIPAC Fellow) — co-developer of ICR and `GraphGP`, NIFTy/NIFTy.re, variational inference; applied ICR to 3D dust mapping with $> 10^8$ DOF.
- **Benjamin Dodge** (Stanford/ISM group) — lead developer of `GraphGP`, GPU-accelerated Vecchia methods.
- **Susan Clark** (Stanford Physics) — PI of the ISM group, `GraphGP` development and scientific applications.
- **Risa Wechsler** (KIPAC/SLAC) — galaxy–halo connection, simulation-based cosmological inference, DESI.

Natural extensions. The project would benefit from additional expertise in several directions: DESI survey systematics and fiber-assignment modelling; Hamiltonian Monte Carlo and variational inference at scale; perturbation-theory–informed kernel design; and connections to the BORG/NIFTy field-level inference communities. Students and postdocs at any career stage who find the ideas here interesting are especially encouraged to reach out—there are well-defined sub-projects at a range of levels, from implementing the first toy-model proof of concept to the theoretical analysis of what the GP marginal likelihood optimally extracts from a Cox process.

Computational resources. `GraphGP` on a single A100 GPU handles $\sim 10^9$ points for generation. Iterative inference on 10^6-galaxy catalogs is well within reach on current hardware available through SLAC and Stanford computing allocations.

14 Timeline

Months 1–3: Implement Poisson–GP Stage 1 with `GraphGP`; test isotropic kernel on 10^4-point toy models; validate field reconstruction and band-power recovery against known truth.

Months 3–6: Implement anisotropic kernel $C_{gg}(r_\mathrm{p}, r_\parallel)$; scale to 10^5–10^6 AbacusSummit mocks; validate $P_{gg,\ell}(k_i)$ recovery and Fisher-matrix covariances against mock-based estimates.

Months 6–9: Integrate DD-kNN as Gaussianity diagnostic; map the non-Gaussianity frontier; implement Stage 2 with multiple theoretical models; first paper on simulation-based validation of the two-stage framework.

Months 9–12: Address DESI-specific systematics (fiber collisions, selection function, light-cone evolution); apply Stage 1 to DESI early data; deliver $P_{gg,\ell}(k_i)$ with covariances for community Stage 2 analyses; second paper.

15 Summary

Field-level cosmological inference from galaxy surveys can be cleanly separated into two stages. Stage 1 uses a Gaussian process with a Poisson likelihood to reconstruct the galaxy density field and learn the anisotropic galaxy power spectrum $P_{gg,\ell}(k)$ as kernel hyperparameters—delivering band-power estimates with internally consistent covariances, all without reference to matter, bias, or cosmology. The multi-output GP extension learns the full matrix of auto- and cross-spectra for galaxy subsamples split by observable properties, turning property-dependent clustering and assembly bias from modelling assumptions into empirical observables. Because derivatives of a GP are themselves GPs, the framework simultaneously delivers the tidal shear tensor and cosmic web classification at every galaxy position, enabling empirical tests of tidal bias and a systematic assessment of how many bias parameters the data require at each scale. Stage 2 takes these spectra, field reconstructions, and local environment characterizations and fits any theoretical model. The `GraphGP` package provides $\mathcal{O}(Nk^3)$ scaling at arbitrary point positions. The 2D DD-kNN statistics provide a stratified Gaussianity diagnostic within Stage 1 and additional non-Gaussian constraints for Stage 2. This architecture—analogous to the CMB's separation of map-making from parameter inference—offers a fundamentally new approach to galaxy survey analysis where measurement and interpretation are cleanly decoupled, where galaxy formation physics becomes an observable rather than a nuisance, and where different theoretical frameworks can be compared on identical data compressions without re-running the expensive field-level inference.

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